Stochastic Finance and Risk Modelling

Chapter 3: One-period model of financial market

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Expanded Study Resource

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Introduction

This document serves as a comprehensive study guide for the course "Stochastic Finance and Risk Modelling". It expands upon the lecture slides relative to Chapter 3: "One-period model of financial market".

In this chapter, we introduce the fundamental concepts of financial modeling in a simplified one-period setting. We define the market, assets (riskless and risky), and portfolios. We then explore the crucial concept of arbitrage and the Fundamental Theorem of Asset Pricing (FTAP), which links the absence of arbitrage to the existence of risk-neutral measures. Finally, we discuss contingent claims (derivatives), their pricing, and market completeness.

Chapter 1

Market, Assets, and Portfolio

1.1 The One-Period Market Model

We consider a simplified model of a financial market that operates over a single time period. The timeline consists of two specific dates:

- $t = 0$: The current time, or "now".
- $t = T = 1$: Accurse future time, or "end of the period".

In this market, there are $d + 1$ assets available for trading. We denote the price of asset i at time t by S_t^i, where $i \in \{0, 1, \ldots, d\}$ and $t \in \{0, 1\}$.

The assets are categorized into two types: a riskless asset and risky assets.

1.1.1 The Riskless Asset

We designate asset 0 as the riskless asset (or risk-free asset). This asset is characterized by a deterministic return over the period.

Definition 1.1 (Riskless Asset)**.** Asset 0 is riskless if its initial price is strictly positive, $S_0^0 > 0$, and its future price is given by:

$$S_1^0 = S_0^0(1 + r), \tag{1.1}$$

where r is a constant known as the *risk-free interest rate*. We assume $r > -1$ to ensure prices remain positive.

Commonly, we normalize the initial price to unit currency, i.e., we choose $S_0^0 = 1$. In this case, $S_1^0 = 1 + r$. This asset can be interpreted as a bank account or a government bond:

- If you invest S_0^0 at time 0, you receive $S_0^0(1 + r)$ at time 1. This corresponds to lending money (saving).
- Conversely, you can borrow S_0^0 at time 0 and repay $S_0^0(1 + r)$ at time 1.

1.1.2 Risky Assets

The remaining d assets, indexed by $i = 1, \ldots, d$, are risky assets. Their prices at time $t = 0$ are known constants, but their prices at time $t = 1$ are uncertain.

We model this uncertainty using a probability space $(\Omega, \mathcal{F}_1, \mathbb{P})$.

- Ω: The sample space, representing the set of all possible states of nature or market scenarios.
- $\mathcal{F}_1$: A σ-algebra of events observable at time 1.
- $\mathbb{P}$: The real-world (or historical) probability measure.

Definition 1.2 (Risky Assets)**.** The initial price vector of the risky assets is a constant vector in $\mathbb{R}^d_+$:

$$S_0 = (S_0^1, \ldots, S_0^d). \tag{1.2}$$

The future price vector at time $t = 1$ is a random vector $S_1 = (S_1^1, \ldots, S_1^d)$ defined on the probability space $(\Omega, \mathcal{F}_1, \mathbb{P})$. Specifically, each S_1^i is a non-negative random variable, meaning it is $\mathcal{F}_1$-measurable and maps $\Omega \to \mathbb{R}_+$.

Construction of the Probability Space

A canonical way to construct this space is to let the sample space be the set of all possible strictly positive price vectors:

$$\Omega := \mathbb{R}^d_+ = (0, \infty)^d. \tag{1.3}$$

The σ-algebra $\mathcal{F}_1$ is the Borel σ-algebra on $\mathbb{R}^d_+$, denoted $\mathcal{B}(\mathbb{R}^d_+)$. The random variable S_1 is then the identity map on Ω:

$$S_1(\omega) := \omega, \quad \forall \omega \in \mathbb{R}^d_+. \tag{1.4}$$

Here, a specific $\omega \in \Omega$ represents a realized scenario of asset prices at time 1.

1.2 Portfolios and Value Processes

An investor participates in the market by constructing a portfolio, which is a collection of assets held between time 0 and time 1.

Definition 1.3 (Portfolio)**.** A portfolio (or trading strategy) is defined by a vector $\phi = (\xi^0, \xi) \in \mathbb{R} \times \mathbb{R}^d$, where:

- $\xi^0 \in \mathbb{R}$ represents the quantity (number of units) of the riskless asset held.
- $\xi = (\xi^1, \ldots, \xi^d) \in \mathbb{R}^d$ represents the quantities of the risky assets held.

Crucially, the coefficients ξ^i are not restricted to be integers or positive.

- $\xi^i > 0$: Long position (buying/holding the asset).
- $\xi^i < 0$: Short position (selling/shorting the asset). Short selling implies borrowing the asset, selling it today, and promising to buy it back at time 1.

1.2.1 Portfolio Value

The value of a portfolio is simply the sum of the values of its constituent assets.

Definition 1.4 (Value Process)**.** Let $\phi = (\xi^0, \xi)$ be a portfolio. The **initial value** V_0^ϕ at time 0 is:

$$V_0^\phi := \xi^0 S_0^0 + \xi \cdot S_0 = \xi^0 S_0^0 + \sum_{i=1}^{d} \xi^i S_0^i. \tag{1.5}$$

The **final value** V_1^ϕ at time 1 is a random variable given by:

$$V_1^\phi := \xi^0 S_1^0 + \xi \cdot S_1 = \xi^0 S_0^0 (1+r) + \sum_{i=1}^{d} \xi^i S_1^i. \tag{1.6}$$

This simple framework allows us to define gains and losses. If $V_1^\phi > V_0^\phi$, the strategy has generated a profit (in nominal terms). However, to properly compare values at different times, we must account for the time value of money, leading to the concept of discounted prices.

Chapter 2

Arbitrage and the Fundamental Theorem of Asset Pricing

2.1 Arbitrage

Arbitrage is a central concept in financial mathematics. Intuitively, an arbitrage opportunity is a trading strategy that offers the possibility of a risk-free profit. In an efficient market, such opportunities should not exist, or at least not persist for long, as rational agents would exploit them until prices adjust.

Definition 2.1 (Arbitrage Opportunity)**.** A portfolio $\phi = (\xi^0, \xi)$ is called an **arbitrage opportunity** if it satisfies the following three conditions:

1. **Zero initial cost**: $V_0^\phi = 0$.
2. **Non-negative final wealth**: $V_1^\phi \geq 0$ almost surely (a.s.) under the probability measure $\mathbb{P}$.
3. **Positive probability of profit**: $\mathbb{P}(V_1^\phi > 0) > 0$.

Condition (1) means the investor starts with nothing. Condition (2) means there is no risk of losing money. Condition (3) means there is a chance to make money. If such a portfolio exists, one could scale it up indefinitely to make infinite profit without risk.

2.1.1 Characterizing Arbitrage without the Risk-Free Asset

We can eliminate the variable ξ^0 using the self-financing condition. If $V_0^\phi = 0$, then:

$$\xi^0 S_0^0 + \xi \cdot S_0 = 0 \implies \xi^0 = -\frac{\xi \cdot S_0}{S_0^0}. \tag{2.1}$$

Substituting this into the expression for V_1^ϕ:

$$\begin{aligned} V_1^\phi &= \xi^0 S_1^0 + \xi \cdot S_1 && (2.2)\\ &= -\frac{\xi \cdot S_0}{S_0^0} S_0^0 (1+r) + \xi \cdot S_1 && (2.3)\\ &= \xi \cdot S_1 - (1+r)\xi \cdot S_0. && (2.4) \end{aligned}$$

This leads to a simplified criterion for arbitrage involving only the risky asset positions ξ.

Proposition 2.1. *The market admits an arbitrage opportunity if and only if there exists a vector $\xi \in \mathbb{R}^d$ such that:*

1. $\xi \cdot S_1 \geq (1+r)\xi \cdot S_0, \quad \mathbb{P}$-*a.s.*
2. $\mathbb{P}(\xi \cdot S_1 > (1+r)\xi \cdot S_0) > 0$.

Interpretation ($d = 1$): For a single risky asset, the condition $\xi \cdot S_1 \geq (1+r)\xi \cdot S_0$ becomes:

- If $\xi > 0$ (long position): $\frac{S_1^1 - S_0^1}{S_0^1} \geq r$. The return of the risky asset dominates the risk-free rate.
- If $\xi < 0$ (short position): $\frac{S_1^1 - S_0^1}{S_0^1} \leq r$. The return of the risky asset is dominated by the risk-free rate.

This implies that if an asset systematically outperforms or underperforms the risk-free rate in all states of the world, an arbitrage exists.

2.2 Discounted Prices and Martingales

To compare cash flows at different times properly, we discount future values back to the present using the risk-free asset as a numeraire.

Definition 2.2 (Discounted Price Process)**.** The discounted price of asset i, denoted by $\tilde{S}_t^i$, is defined as:

$$\tilde{S}_t^i = \frac{S_t^i}{S_t^0}, \quad \text{for } t \in \{0, 1\}. \tag{2.5}$$

Clearly, $\tilde{S}_0^i = S_0^i / S_0^0$ and $\tilde{S}_1^i = S_1^i / [S_0^0(1+r)]$. For the riskless asset itself, $\tilde{S}_t^0 = 1$ for all t.

Intuition behind the Discounted Price: The discounted price can be understood through three key perspectives:

1. **Change of Numéraire (Measuring in "Savings Units"):** Instead of measuring the price of a risky asset in currency (e.g., dollars), we measure it in units of the risk-free asset. The discounted price $\tilde{S}_t^i$ answers the question: "How many units of the risk-free asset S^0 do I need to sell to buy one unit of the risky asset S^i?"
2. **Isolating the Risk Premium:** A risk-free investment automatically grows by a factor of $(1+r)$. If a risky asset also grows by exactly $(1+r)$, it offers no additional return for the risk taken. By dividing by S_1^0, we remove this "baseline" growth.
   - If $\tilde{S}_1^i > \tilde{S}_0^i$, the asset outperformed the risk-free rate.
   - If $\tilde{S}_1^i = \tilde{S}_0^i$, the asset exactly matched the risk-free rate.
   - If $\tilde{S}_1^i < \tilde{S}_0^i$, the asset underperformed the risk-free rate.
3. **Time Value of Money:** To fairly compare a future value S_1^i with a present value S_0^i, we must bring the future value to the present terms. Dividing by $(1+r)$ "discounts" the future price, accounting for the opportunity cost of capital.

Similarly, the discounted value of a portfolio is $\tilde{V}_t^\phi = V_t^\phi / S_t^0$. The change in discounted value is determined solely by the change in discounted risky asset prices, $\Delta\tilde{S}_1 = \tilde{S}_1 - \tilde{S}_0$:

$$\tilde{V}_1^\phi - \tilde{V}_0^\phi = \xi \cdot (\tilde{S}_1 - \tilde{S}_0) = \xi \cdot \Delta\tilde{S}_1. \tag{2.6}$$

This allows us to restate the arbitrage condition even more compactly:

Proposition 2.2. *An arbitrage opportunity exists iff there is a $\xi \in \mathbb{R}^d$ such that $\xi \cdot \Delta\tilde{S}_1 \geq 0$ a.s. and $\mathbb{P}(\xi \cdot \Delta\tilde{S}_1 > 0) > 0$.*

2.3 Risk-Neutral Measure (Martingale Measure)

Definition 2.3 (Martingale Measure)**.** A probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_1)$ is called a **risk-neutral measure** or **martingale measure** if:

1. $\mathbb{Q}$ is equivalent to $\mathbb{P}$ (denoted $\mathbb{Q} \sim \mathbb{P}$), meaning they share the same null sets: $\mathbb{P}(A) = 0 \iff \mathbb{Q}(A) = 0$.

2. The discounted price process is a martingale under $\mathbb{Q}$. In our one-period setting, this means:
$$\tilde{S}_0^i = \mathbb{E}^{\mathbb{Q}}[\tilde{S}_1^i], \quad \forall i = 1, \ldots, d. \tag{2.7}$$

Using the definition of discounted prices, the martingale condition can be written as:

$$S_0^i = \mathbb{E}^{\mathbb{Q}}\left[\frac{S_1^i}{1+r}\right] \implies S_0^i(1+r) = \mathbb{E}^{\mathbb{Q}}[S_1^i]. \tag{2.8}$$

This states that under the risk-neutral measure $\mathbb{Q}$, the expected return of every asset equals the risk-free rate r. The "risk premium" vanishes, hence the term "risk-neutral".

2.4 Fundamental Theorem of Asset Pricing (FTAP)

The FTAP is the cornerstone of modern finance, establishing a deep link between the economic concept of no-arbitrage and the mathematical concept of martingales.

Theorem 2.3 (First Fundamental Theorem of Asset Pricing)**.** *The market is arbitrage-free if and only if there exists at least one risk-neutral martingale measure $\mathbb{Q}$ equivalent to $\mathbb{P}$.*

Sketch of Idea. The "if" direction is relatively straightforward. Suppose such a $\mathbb{Q}$ exists. Then for any portfolio with $V_0^\phi = 0$, the discounted value process is a martingale (linear combination of martingales). Thus $\mathbb{E}^{\mathbb{Q}}[\tilde{V}_1^\phi] = \tilde{V}_0^\phi = 0$. If $\tilde{V}_1^\phi \geq 0$ a.s., then its expectation can only be 0 if $\tilde{V}_1^\phi = 0$ a.s. This contradicts the requirement for strict profit with positive probability. Hence, no arbitrage exists. The "only if" direction requires the Separating Hyperplane Theorem from convex analysis and is more involved. □

2.5 Portfolio Returns

We define the return of a portfolio ϕ (with $V_0^\phi \neq 0$) as:

$$R(\phi) = \frac{V_1^\phi - V_0^\phi}{V_0^\phi}. \tag{2.9}$$

Under any risk-neutral measure $\mathbb{Q}$, the expected return of *any* portfolio is equal to the risk-free rate r.

$$\mathbb{E}^{\mathbb{Q}}[R(\phi)] = r. \tag{2.10}$$

This is a powerful result: under $\mathbb{Q}$, it doesn't matter how risky a portfolio is; its expected appreciation is simply the risk-free rate.

Chapter 3

Derivatives and Pricing

3.1 Payoffs and Replicable Payoffs

The primary goal of financial engineering is to price and hedge financial products. These products are often defined by their future cash flows, which depend on the uncertain state of the market.

Definition 3.1 (Payoff)**.** A **contingent claim** or **derivative** is defined by its payoff at maturity $T = 1$. Mathematically, a payoff is a non-negative random variable H on the probability space $(\Omega, \mathcal{F}_1, \mathbb{P})$.

Examples include options, futures, and swaps. The key question is: "What is the fair price of H at time $t = 0$?"

3.1.1 Replicable Payoffs

One way to price a derivative is through replication. If we can construct a portfolio that exactly matches the payoff H in every possible state of the world, then the price of H today must be the initial cost of that portfolio (assuming no arbitrage).

Definition 3.2 (Replicable Payoff)**.** A payoff H is said to be **replicable** (or specific) if there exists a portfolio $\phi = (\xi^0, \xi)$ such that its final value equals H almost surely:

$$V_1^\phi = \xi^0 S_1^0 + \xi \cdot S_1 = H, \quad \mathbb{P}\text{-a.s.} \tag{3.1}$$

Such a portfolio is called a **replicating portfolio** or a **hedge**.

The set of all replicable payoffs is a linear subspace of the space of random variables, given by:

$$\mathcal{H}_{rep} = \left\{ \xi^0 S_1^0 + \xi \cdot S_1 \mid (\xi^0, \xi) \in \mathbb{R}^{d+1} \right\}. \tag{3.2}$$

Proposition 3.1 (Law of One Price)**.** *In an arbitrage-free market, if a payoff H is replicable, its price is unique. Specifically, if two portfolios ϕ and ψ both replicate H (i.e., $V_1^\phi = V_1^\psi = H$), then they must have the same initial cost:*

$$V_0^\phi = V_0^\psi. \tag{3.3}$$

Proof. Consider the portfolio $\phi - \psi$. $V_1^{\phi-\psi} = V_1^\phi - V_1^\psi = H - H = 0$. If $V_0^{\phi-\psi} < 0$, we have an arbitrage (immediate profit with no future liability). If $V_0^{\phi-\psi} > 0$, the reverse portfolio $\psi - \phi$ is an arbitrage. Thus $V_0^{\phi-\psi} = 0$. □

3.2 Common Types of Derivatives

Most derivatives are functions of the underlying asset prices at maturity. Let $\mathcal{F}_1 = \sigma(S_1^0, \ldots, S_1^d)$. A derivative is often of the form $H = f(S_1^0, \ldots, S_1^d)$ for some measurable function $f : \mathbb{R}^{d+1} \to \mathbb{R}$.

3.2.1 Forward Contract

A forward contract is an agreement to buy an asset S^i at time $T = 1$ for a fixed delivery price K.

- **Payoff**: $H^F = S_1^i - K$.
- Can be positive or negative.
- **Replication**: Buy 1 unit of asset i and borrow $K/(1+r)$ at time 0. At time 1, you own the asset S_1^i and owe K. Total value: $S_1^i - K$. Initial cost: $S_0^i - \frac{K}{1+r}$. This is the arbitrage-free price.

3.2.2 Call Option

A call option gives the holder the right (but not the obligation) to buy an asset S^i at strike price K.

- The holder exercises the option only if $S_1^i > K$.
- **Payoff**: $H^C = \max(S_1^i - K, 0) = (S_1^i - K)^+$.
- Strictly non-negative.

3.2.3 Put Option

A put option gives the holder the right to sell an asset S^i at strike price K.

- The holder exercises the option only if $S_1^i < K$.
- **Payoff**: $H^P = \max(K - S_1^i, 0) = (K - S_1^i)^+$.

3.3 Arbitrage-Free Pricing

What if a payoff is not replicable? Can we still assign it a price? We define the price p of a derivative H such that introducing this new asset into the market does not create an arbitrage opportunity.

Definition 3.3 (Arbitrage-Free Price)**.** Let $\mathcal{M}$ be the original market with assets $(S^0, S^1, \ldots, S^d)$. Consider a derivative H. We call p an **arbitrage-free price** for H if the extended market $\mathcal{N} = \mathcal{M} \cup \{S^{d+1}\}$ with:

$$S_0^{d+1} = p, \quad S_1^{d+1} = H \tag{3.4}$$

remains arbitrage-free.

Theorem 3.2 (Pricing Formula)**.** *Assume the market $\mathcal{M}$ is arbitrage-free. The set of all possible arbitrage-free prices for a derivative H is given by the set of expected discounted payoffs under all risk-neutral measures $\mathbb{Q}$:*

$$\mathcal{P}(H) = \left\{ \mathbb{E}^{\mathbb{Q}} \left[\frac{H}{S_1^0} \right] \mid \mathbb{Q} \sim \mathbb{P} \textit{ is a martingale measure} \right\}. \tag{3.5}$$

This connects pricing directly to the measures $\mathbb{Q}$.

- If H is replicable, its price is unique and equals $\mathbb{E}^{\mathbb{Q}}[\frac{H}{S_1^0}]$ for *any* martingale measure $\mathbb{Q}$.
- If H is not replicable, there may be a range of valid prices, corresponding to different views on risk (different $\mathbb{Q}$s).

This leads to the concept of market completeness.

Chapter 4

Complete Markets

We now address the question of replicability for all possible payoffs. If every derivative can be perfectly hedged, the market is "complete". This is a highly desirable property, as it implies that any financial risk can be eliminated through trading.

4.1 Definition and Characterization

Definition 4.1 (Complete Market)**.** A financial market $\mathcal{M}$ is said to be **complete** if every payoff $H \in L^0(\Omega, \mathcal{F}_1, \mathbb{P})$ is replicable. That is, for any random variable H, there exists a portfolio $\phi = (\xi^0, \xi)$ such that $V_1^\phi = H$ almost surely.

In a complete market, every risk is "insurable" or "hedgeable".

4.1.1 Connection to Martingale Measures

The completeness of a market is intimately related to the uniqueness of the risk-neutral measure.

Theorem 4.1 (Second Fundamental Theorem of Asset Pricing)**.** *Consider an arbitrage-free market model. The market is complete if and only if there exists a* ***unique*** *martingale measure $\mathbb{Q}$ equivalent to $\mathbb{P}$.*

This theorem provides a powerful tool to check for completeness without explicitly constructing hedging strategies for every payoff. We just need to check the set of martingale measures:

- **No** $\mathbb{Q}$: Arbitrage exists. The model is flawed.
- **Unique** $\mathbb{Q}$: The market is arbitrage-free and complete. Every derivative has a unique price. This is the ideal theoretical setting (e.g., Black-Scholes model in continuous time).
- **Multiple** $\mathbb{Q}$**s**: The market is arbitrage-free but incomplete. Some derivatives cannot be replicated perfectly. Their prices are not unique but lie within an interval determined by the range of expectations under different $\mathbb{Q}$s.

4.2 Geometric Interpretation

In our finite probability space setting with $|\Omega| = K$ states:

- The space of all payoffs is isomorphic to $\mathbb{R}^K$.
- The space of replicable payoffs is the subspace spanned by the asset price vectors (a linear subspace of dimension at most $d + 1$).
- Completeness means this subspace spans the entire $\mathbb{R}^K$.
- Thus, a necessary condition for completeness is that the number of linearly independent assets $(d + 1)$ must be at least the number of states (K).

Typically, we need as many risky assets as there are sources of uncertainty/risk factors to complete the market.

4.3 Summary of Key Results

Economic Concept	**Mathematical Equivalent (Martingale Theory)**
No Arbitrage	Existence of at least one EMM ($\mathbb{Q} \sim \mathbb{P}$)
Completeness	Uniqueness of the EMM ($\mathbb{Q}$)
Arbitrage-Free Price	Expectation under an EMM: $\mathbb{E}^{\mathbb{Q}}[\tilde{H}]$
Replication Strategy	Representation in Martingale Representation Theorem

Table 4.1: Correspondence between Finance and Mathematics

This chapter has established the foundational one-period framework. Subsequent chapters will extend these ideas to multi-period models and, eventually, continuous-time models, where the same fundamental principles—no arbitrage and risk-neutral pricing—remain the guiding lights.